

On the Primitivity and Diameter of Modular Syracuse Graphs: Exact Structural Properties and the De Bruijn Graph Correspondence

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Abstract

This document presents a complete structural analysis of the modular Syracuse graph G_p defined on the finite ring $\mathbb{Z}/2^p\mathbb{Z}$. We rigorously demonstrate that this graph is obtained from the binary De Bruijn graph $B(2, p)$ by a local path correspondence (rerouting) under a right-shift convention [2]. Leveraging this algebraic bridge, we prove that G_p is strongly connected, that its diameter is exactly equal to $2p - 1$, and that its primitivity exponent is exactly equal to $2p$ [1].

Framing Note: These results concern finite modular systems only and should not be interpreted as a proof of the classical Collatz conjecture on $\mathbb{N}$. In accordance with the limitations discussed in section 6, the loss of binary branching in the infinite limit prevents any direct extension of these finite-state properties to the global dynamics on $\mathbb{N}$ [3].

1 Definitions and Mathematical Framework

1.1 Historical Context

The study of Collatz-type dynamical systems dates back to the work of Everett [5] and the subsequent generalizations by Lagarias [4]. The modular approach, where the dynamics are considered modulo powers of 2, was developed by Bernstein and Lagarias [1], who introduced a conjugacy between the Collatz map and the binary shift on the ring of 2-adic integers. More recently, Laarhoven and de Weger [2] established the connection between the modular Collatz graph and De Bruijn graphs. While the isomorphism between $C(p)$ and $B(2, p)$ is known from [2], the exact metric properties of the associated graph G_p — namely its diameter and primitivity exponent — have not been previously determined. The present work fills this gap by deriving explicit formulas for these quantities, showing that the local rerouting induced by the Syracuse map preserves the De Bruijn structure in a quantitatively precise way.

1.2 The modular Syracuse graph G_p

For any integer $p \geq 2$, let $V_p = \mathbb{Z}/2^p\mathbb{Z} = \{0, 1, \dots, 2^p - 1\}$. The modular Syracuse graph $G_p = (V_p, E_p)$ is a directed graph whose transitions are defined as follows:

- **If n is even** ($n \equiv 0 \pmod{2}$), then n has exactly two successors:

$$s_1(n) = \frac{n}{2} \pmod{2^p}, \quad s_2(n) = \frac{n}{2} + 2^{p-1} \pmod{2^p}.$$

- **If n is odd** ($n \equiv 1 \pmod{2}$), then n has a unique successor:

$$s(n) = 3n + 1 \pmod{2^p}.$$

1.3 The modular binary Collatz graph $C(p)$

To analyze the structure of G_p , we introduce the standard modular binary Collatz graph $C(p) = (V_p, E'_p)$, as studied by Laarhoven and de Weger (2013) [2]. In $C(p)$, every vertex $n \in V_p$ has exactly two successors:

- **If n is even:**

$$\mathcal{T}_C(n) = \left\{ \frac{n}{2} \pmod{2^p}, \frac{n}{2} + 2^{p-1} \pmod{2^p} \right\}.$$

- **If n is odd:**

$$\mathcal{T}_C(n) = \left\{ \frac{3n+1}{2} \pmod{2^p}, \frac{3n+1}{2} + 2^{p-1} \pmod{2^p} \right\}.$$

1.4 The binary De Bruijn graph $B(2, p)$

The binary De Bruijn graph $B(2, p)$ is a regular directed graph with 2^p vertices, where each vertex is represented by a binary word of length p : $\mathbf{x} = (x_p, x_{p-1}, \dots, x_1) \in \{0, 1\}^p$, where x_1 represents the least significant bit (LSB) and x_p represents the most significant bit (MSB).

The outgoing edges from a vertex $\mathbf{x}$ are defined by a **right-shift** operation (where the LSB x_1 is discarded, the remaining bits are shifted to the right, and a new bit $b \in \{0, 1\}$ is inserted at the leftmost MSB position) [2]:

$$\mathbf{y}_b = (b, x_p, x_{p-1}, \dots, x_2).$$

Each vertex has an out-degree equal to 2, corresponding to the choices $b = 0$ and $b = 1$.

Consequently, the isomorphism between $C(p)$ and $B(2, p)$ is given by the natural binary representation of each integer $n \in \mathbb{Z}/2^p\mathbb{Z}$.

1.5 The Bernstein-Lagarias Conjugacy Map

For any $p \geq 2$, we define the conjugacy map $\Phi_p : \mathbb{Z}/2^p\mathbb{Z} \rightarrow \{0, 1\}^p$. For any $n \in \mathbb{Z}/2^p\mathbb{Z}$ [1]:

$$\Phi_p(n) = (x_p(n), x_{p-1}(n), \dots, x_1(n)),$$

where $x_i(n) = T^{i-1}(n) \pmod{2}$, with $T(k) = k/2$ if k is even and $T(k) = (3k+1)/2$ if k is odd.

This map Φ_p is a bijection, known as the **Bernstein-Lagarias conjugacy map** [1]. Moreover, it defines a rigorous **graph isomorphism** between the modular Collatz graph $C(p)$ and the De Bruijn graph $B(2, p)$ [2]. Indeed, for $m = T(n) + b 2^{p-1}$ (the two successors of n in $C(p)$), one verifies directly that

$$\Phi_p(m) = (b, x_p(n), x_{p-1}(n), \dots, x_2(n)),$$

which is precisely the right-shift edge rule of $B(2, p)$.

Under this isomorphism, the parity of a vertex $n \in C(p)$ corresponds exactly to the least significant bit (LSB) of the binary vector $\Phi_p(n)$, namely $x_1(n)$ (the rightmost bit).

2 Structural Path Correspondence and Exact Rerouting

Theorem 2.1 (Structural Path Correspondence Theorem). *The graph G_p is structurally obtained from the binary Collatz graph $C(p)$ (and hence from the De Bruijn graph $B(2, p)$) by a local rerouting: for each odd vertex n , the two outgoing edges are replaced by a path of length 2 passing through the pre-existing even vertex $3n + 1$. Consequently, every edge of $C(p)$ emanating from an odd vertex corresponds uniquely to a directed path of length exactly 2 in G_p , while edges emanating from even vertices are preserved identically.*

Proof. To prove this theorem, we must show that for any odd vertex $n \in V_p$, the path of length 2 in G_p starting from n reaches exactly the same set of vertices as the single-step transitions starting from n in $C(p)$.

Let $n \in V_p$ be an odd number. In $C(p)$, the set of direct targets from n is:

$$\mathcal{T}_C(n) = \left\{ \frac{3n+1}{2} \bmod 2^p, \left(\frac{3n+1}{2} + 2^{p-1} \right) \bmod 2^p \right\}.$$

In G_p , the first transition from n (which is odd) must lead to its unique successor:

$$y = (3n + 1) \bmod 2^p.$$

Since n is odd, $3n + 1$ is even, which implies that its remainder modulo 2^p , namely y , is also an even number. Since y is even, the even transition rule of G_p applies. The two successors of y in G_p are:

$$\mathcal{T}_G(n) = \left\{ \frac{y}{2} \bmod 2^p, \left(\frac{y}{2} + 2^{p-1} \right) \bmod 2^p \right\}.$$

By the definition of division, we can write the following Euclidean division on integers:

$$3n + 1 = q \cdot 2^p + y \quad \text{with } 0 \leq y < 2^p.$$

Since y is even, we can divide this equation by 2:

$$\frac{3n+1}{2} = q \cdot 2^{p-1} + \frac{y}{2}.$$

Since $0 \leq y < 2^p$, we have $0 \leq \frac{y}{2} < 2^{p-1}$. Consequently, $\frac{y}{2}$ is exactly the remainder of the Euclidean division of $\frac{3n+1}{2}$ by 2^{p-1} :

$$\frac{y}{2} = \frac{3n+1}{2} \bmod 2^{p-1}.$$

Now, let us examine the value of $t_{C1} = \frac{3n+1}{2} \bmod 2^p$. This integer must be congruent to $\frac{3n+1}{2}$ modulo 2^{p-1} . Within the interval $[0, 2^p - 1]$, there exist only two values satisfying this congruence:

- **Either** $t_{C1} = \frac{y}{2}$
- **Or** $t_{C1} = \frac{y}{2} + 2^{p-1}$

Two mutually exclusive cases then arise depending on the parity of the quotient q :

1. **Case where q is even** ($q = 2k$, with $k \in \mathbb{Z}$): The equation becomes:

$$\frac{3n+1}{2} = k \cdot 2^p + \frac{y}{2} \implies t_{C1} = \frac{3n+1}{2} \bmod 2^p = \frac{y}{2}.$$

The second successor in $C(p)$ is then:

$$t_{C2} = \left(\frac{3n+1}{2} + 2^{p-1} \right) \bmod 2^p = \frac{y}{2} + 2^{p-1}.$$

2. **Case where q is odd** ($q = 2k + 1$, with $k \in \mathbb{Z}$): The equation becomes:

$$\frac{3n+1}{2} = k \cdot 2^p + 2^{p-1} + \frac{y}{2} \implies t_{C1} = \frac{3n+1}{2} \bmod 2^p = \frac{y}{2} + 2^{p-1}.$$

The second successor in $C(p)$ is then:

$$t_{C2} = \left(\frac{3n+1}{2} + 2^{p-1} \right) \bmod 2^p = \left(\frac{y}{2} + 2^{p-1} + 2^{p-1} \right) \bmod 2^p = \frac{y}{2}.$$

In both cases, the set of targets reached after two steps in G_p is strictly identical to the set of targets reached in a single step in $C(p)$:

$$\mathcal{T}_C(n) = \mathcal{T}_G(n) = \left\{ \frac{y}{2} \bmod 2^p, \left(\frac{y}{2} + 2^{p-1} \right) \bmod 2^p \right\}.$$

This demonstrates the exactness of the path correspondence. The graph G_p preserves the path structure of the De Bruijn graph, except that each transition originating from an odd state has an intermediate state $y = 3n + 1 \pmod{2^p}$ inserted along its path. $\square$

3 Structural Properties and Strong Connectivity

Theorem 3.1 (Strong Connectivity Theorem). *The graph G_p is strongly connected for any $p \geq 2$.*

Proof. To prove that G_p is strongly connected, it is sufficient to establish two conditions:

1. **General reachability from 0:** From vertex 0, any vertex $j \in V_p$ can be reached in at most p steps.
2. **Systematic return to 0:** From any vertex $i \in V_p$, vertex 0 can be reached in a finite number of steps.

Proof of Condition 1 (From 0 to any vertex) We prove by induction on k (for $1 \leq k \leq p$) that the set of vertices reachable from 0 in exactly k steps in G_p , denoted by $\mathcal{A}_k(0)$, contains all multiples of 2^{p-k} in $\mathbb{Z}/2^p\mathbb{Z}$.

- **Base case** ($k = 1$): Vertex 0 is even. Its direct successors in G_p are:

$$s_1(0) = \frac{0}{2} = 0 \quad \text{and} \quad s_2(0) = \frac{0}{2} + 2^{p-1} = 2^{p-1}.$$

The set of vertices reached in 1 step is $\mathcal{A}_1(0) = \{0, 2^{p-1}\}$, which corresponds exactly to all multiples of 2^{p-1} in $\mathbb{Z}/2^p\mathbb{Z}$.

- **Inductive step:** Assume that for a given integer k ($1 \leq k < p$), the set $\mathcal{A}_k(0)$ contains all multiples of 2^{p-k} . We now show that the next layer $\mathcal{A}_{k+1}(0)$ contains all multiples of the finer modulus $2^{p-(k+1)}$. To this end, let m be a multiple of $2^{p-(k+1)}$. Two cases arise:

- **If m is a multiple of 2^{p-k} :** By the induction hypothesis, $m \in \mathcal{A}_k(0)$. Since $0 \in \mathcal{A}_1(0)$ and $0 \rightarrow 0$ is a self-loop, we can extend the path from 0 to m by one additional step by looping on 0 at the start. Thus, $m \in \mathcal{A}_{k+1}(0)$.
- **If m is not a multiple of 2^{p-k} :** Then m can be written as $m = d \cdot 2^{p-(k+1)}$ where d is an odd integer. Consider the vertex $n = d \cdot 2^{p-k}$. Since $k < p$, the exponent $p - k \geq 1$, which guarantees that n is an even number. By the induction hypothesis, $n \in \mathcal{A}_k(0)$. Its successors in G_p are:

$$s_1(n) = \frac{d \cdot 2^{p-k}}{2} = d \cdot 2^{p-(k+1)} = m,$$

$$s_2(n) = d \cdot 2^{p-(k+1)} + 2^{p-1}.$$

Since $n \in \mathcal{A}_k(0)$ and the transition $n \rightarrow m$ is valid, we have $m \in \mathcal{A}_{k+1}(0)$.

The inductive step is complete. Consequently, for $k = p$, the set $\mathcal{A}_p(0)$ contains all multiples of $2^{p-p} = 2^0 = 1$, which corresponds to the entirety of the set V_p .

Proof of Condition 2 (From any vertex to 0) Let $i \in V_p$ be any vertex. By the Structural Path Correspondence Theorem (theorem 2.1), every directed edge of the standard Collatz graph $C(p)$ corresponds to a directed path in G_p : the correspondence is of length 1 if the source vertex is even, and of length 2 if the source vertex is odd (passing through the intermediate vertex $3n + 1$).

Since $C(p)$ is isomorphic to the De Bruijn graph $B(2, p)$ (see section 1.4), and since $B(2, p)$ is strongly connected (the all-zero word $(0, 0, \dots, 0)$ is reachable from any word by injecting only zeros after at most p shifts), there exists a directed path from i to 0 in $C(p)$.

Applying the path correspondence of section 2 to each edge of this path yields a directed walk from i to 0 in G_p . Hence, every vertex in V_p can reach 0.

Combining Condition 1 (every vertex is reachable from 0) and Condition 2 (every vertex can reach 0), we conclude that G_p is strongly connected. $\square$

4 Diameter Theorem

Theorem 4.1 (Exact Diameter Theorem). *For any $p \geq 2$, the diameter of the modular Syracuse graph G_p is exactly:*

$$\text{diam}(G_p) = 2p - 1.$$

Proof. 1. Upper Bound ($\text{diam}(G_p) \leq 2p - 1$). Let $u, v \in V_p$ be arbitrary. Since $C(p)$ is isomorphic to the De Bruijn graph $B(2, p)$ (section 1.4), there exists a directed path in $C(p)$ from u to v of length $L \leq p$. Let this path be

$$u = n_0 \rightarrow n_1 \rightarrow \dots \rightarrow n_L = v.$$

By the Structural Path Correspondence Theorem (theorem 2.1), each transition $n_i \rightarrow n_{i+1}$ in $C(p)$ corresponds to a path in G_p of length 1 if n_i is even, and length 2 if n_i is odd. Hence,

$$\text{dist}_{G_p}(u, v) \leq L + |\{i \in \{0, \dots, L-1\} : n_i \text{ is odd}\}|.$$

If $L \leq p-1$, then this bound is at most $2L \leq 2p-2 < 2p-1$. So suppose $L = p$.

If among the p sources $n_0, \dots, n_{p-1}$ at least one is even, the total length is at most

$$1 \cdot (\text{one even step}) + 2 \cdot (p-1) = 2p-1.$$

It remains to examine the critical case where $n_0, \dots, n_{p-1}$ are all odd. We first show that this forces $u = 2^p - 1$. Indeed, since each n_k is odd, its LSB is 1. By the Bernstein–Lagarias conjugacy (section 1.4), the LSB of $n_k = T^k(u)$ is exactly the $(k+1)$ -th bit of $\Phi_p(u)$. Thus, all p bits of $\Phi_p(u)$ are 1, so $\Phi_p(u) = (1, 1, \dots, 1)_2$ and hence $u = 2^p - 1$.

We now prove directly, by explicit construction, that for $u = 2^p - 1$, the distance to any $v \in V_p$ is at most $2p-1$. This construction does not rely on the geodesic of $C(p)$ chosen above; it is an independent argument.

- **Case v even.** Write $v \equiv 2m \pmod{2^p}$. Since 3 is invertible modulo 2^p , there exists a unique $w \in \mathbb{Z}/2^p\mathbb{Z}$ such that $3w + 1 \equiv v \pmod{2^p}$. Since v is even, $v-1$ is odd, and $3^{-1} \pmod{2^p}$ is odd (as $3^{-1} \equiv 3^{2^{p-2}-1} \pmod{2^p}$ for $p \geq 2$), their product w is odd. In the De Bruijn graph $C(p)$, there exists a path of length $p-1$ from $u = (1, \dots, 1)_2$ to $w = (b_{p-1}, \dots, b_1, 1)_2$, obtained by injecting successively the bits $b_1, b_2, \dots, b_{p-1}$:

$$(1, 1, \dots, 1) \xrightarrow{b_1} (b_1, 1, \dots, 1) \xrightarrow{b_2} (b_2, b_1, 1, \dots, 1) \xrightarrow{b_3} \dots \xrightarrow{b_{p-1}} (b_{p-1}, \dots, b_1, 1) = w.$$

All $p-1$ sources of this path are odd because the LSB during the first $p-1$ shifts is always one of the original 1s. Hence, in G_p , this path has length $2(p-1)$. The final transition $w \rightarrow 3w + 1 = v$ is a single edge in G_p (since w is odd). Total length:

$$2(p-1) + 1 = 2p-1.$$

- **Case v odd.** Let $\Phi_p(v) = (x_p, \dots, x_2, 1)_2$. In $C(p)$, the path from u to v of length $p-1$ is obtained by injecting the bits $b_1 = x_2, b_2 = x_3, \dots, b_{p-1} = x_p$:

$$(1, 1, \dots, 1) \xrightarrow{x_2} (x_2, 1, \dots, 1) \xrightarrow{x_3} (x_3, x_2, 1, \dots, 1) \xrightarrow{x_4} \dots \xrightarrow{x_p} (x_p, \dots, x_2, 1) = v.$$

Since all $p-1$ sources of this path are odd, the corresponding path in G_p has length exactly $2(p-1) = 2p-2$, which is strictly less than $2p-1$.

Thus, in all cases, for every pair $u, v \in V_p$, we have $\text{dist}_{G_p}(u, v) \leq 2p-1$.

2. Lower Bound ($\text{diam}(G_p) \geq 2p-1$). We exhibit a pair of vertices whose distance is exactly $2p-1$. Consider

$$u = 2^p - 1, \quad v = 0.$$

Under the De Bruijn isomorphism, $\Phi_p(u) = (1, 1, \dots, 1)$ and $\Phi_p(v) = (0, 0, \dots, 0)$.

The unique shortest path in $B(2, p)$ from u to v has length p and consists of injecting p successive zeros. The first $p-1$ transitions originate from odd vertices and hence cost 2 steps each in G_p ; the final transition $x \rightarrow 0$ costs 1 step. This yields a path of length $2(p-1) + 1 = 2p-1$.

To see that no shorter path exists, we use the following lemma.

Lemma 4.2. *Every walk in $C(p)$ from $(1, 1, \dots, 1)$ to $(0, 0, \dots, 0)$ contains at least p transitions whose source vertex is odd.*

Proof of the lemma. Mark the p initial bits of the word $(1, 1, \dots, 1)$. Each marked bit occupies some position in the register. At every De Bruijn shift, exactly one bit is discarded, namely the current least significant bit (LSB). A marked bit, once discarded, can never re-enter the register, because new bits are always injected on the left while the register shifts right. Therefore, to reach the all-zero word $(0, 0, \dots, 0)$, every one of the p marked bits must be discarded at least once during the walk.

Whenever a marked bit is discarded, that bit is the current LSB and is equal to 1. Hence the source vertex of that De Bruijn transition is odd. Since the p marked bits must be discarded at least once, the walk contains at least p transitions whose source is odd. $\square$

Now consider any walk from u to 0 in G_p . By the Structural Path Correspondence (theorem 2.1), it corresponds to a walk in $C(p)$ from $(1, \dots, 1)$ to $(0, \dots, 0)$. By theorem 4.2, this underlying walk contains at least p transitions originating from odd vertices.

In G_p , each such odd-origin transition costs 2 steps, except possibly one: the special terminal shortcut $x \rightarrow 0$ which occurs when $3x + 1 \equiv 0 \pmod{2^p}$ and costs only 1 step. Since this shortcut necessarily lands at 0, it can occur at most once at the very end of any shortest path. Therefore, any walk from u to 0 in G_p has length at least

$$2(p - 1) + 1 = 2p - 1.$$

Since we have already constructed a path of length exactly $2p - 1$, we conclude that

$$\text{dist}_{G_p}(u, 0) = 2p - 1.$$

Combining the upper and lower bounds yields $\text{diam}(G_p) = 2p - 1$. $\square$

5 Primitivity Exponent Theorem

Theorem 5.1 (Exact Primitivity Exponent Theorem). *For any $p \geq 2$, the primitivity exponent of the modular Syracuse graph G_p is exactly:*

$$\exp(G_p) = 2p.$$

Proof. 1. Exact Length Formula for Walks in G_p . Let $u, v \in V_p$ have De Bruijn representations:

$$\Phi_p(u) = (a_p, a_{p-1}, \dots, a_1)_2, \quad \Phi_p(v) = (y_p, y_{p-1}, \dots, y_1)_2.$$

A walk of length $L \geq p$ from u to v in $C(p) \cong B(2, p)$ is uniquely determined by the sequence of L injected bits $c_1, c_2, \dots, c_L$. To end at v , the last p injected bits must satisfy

$$c_{L-p+k} = y_{p-k+1} \quad \text{for } 1 \leq k \leq p.$$

The first $L - p$ bits, denoted $b_1, \dots, b_{L-p}$ (where $b_k = c_k$), are free.

By the Structural Path Correspondence Theorem (theorem 2.1), the length of the corresponding walk in G_p equals L plus the number of transitions in $C(p)$ that originate from an odd vertex. A vertex w_k along the De Bruijn path is odd iff its least significant bit (LSB) is 1.

For the first p transitions ($0 \leq k \leq p-1$), the bit shifted out (which becomes the parity of w_k) is a_{k+1} , a bit of the initial state u .

For subsequent transitions ($p \leq k \leq L-1$), the bit shifted out is b_{k-p+1} , a free injected bit.

Therefore,

$$\sum_{k=0}^{L-1} \mathbb{1}_{\{w_k \text{ odd}\}} = \sum_{j=1}^p a_j + \sum_{k=1}^{L-p} b_k = H(\Phi_p(u)) + S,$$

where $H(\cdot)$ denotes the Hamming weight and $S = \sum_{k=1}^{L-p} b_k$.

Thus, the exact length of the corresponding walk in G_p is

$$\ell_{G_p}(u, v) = L + H(\Phi_p(u)) + S.$$

2. Upper Bound ($\exp(G_p) \leq 2p$).

We begin with a standard observation about the De Bruijn graph.

Lemma 5.2. *For every vertex u of $B(2, p)$, every integer $L \geq p$, and every binary sequence*

$$(c_1, \dots, c_L) \in \{0, 1\}^L,$$

there exists a unique walk of length L starting at u whose successive edge labels are $c_1, \dots, c_L$.

Proof. At each step, the De Bruijn graph has exactly one outgoing edge labeled 0 and one outgoing edge labeled 1. Therefore the sequence of labels determines uniquely the successive vertices of the walk. $\square$

We must show that for any $u, v \in V_p$, there exists a walk of length exactly $2p$ in G_p .

Let $h = H(\Phi_p(u))$. We need to find integers $L \geq p$ and S with $0 \leq S \leq L - p$ such that

$$L + h + S = 2p \iff S = 2p - L - h.$$

The constraint $0 \leq S \leq L - p$ becomes:

$$0 \leq 2p - L - h \leq L - p.$$

This is equivalent to the following two inequalities:

$$L \leq 2p - h, \tag{1}$$

$$L \geq \left\lceil \frac{3p - h}{2} \right\rceil. \tag{2}$$

Since $0 \leq h \leq p$, we have:

$$\left\lceil \frac{3p - h}{2} \right\rceil \leq \left\lceil \frac{3p}{2} \right\rceil \leq 2p - h \quad \text{for all } h \leq p.$$

Furthermore, the lower bound is at least p (since $3p - h \geq 2p$), so $L \geq p$ is automatically satisfied. Therefore, there always exists an integer L in the interval

$$\left[\left\lceil \frac{3p - h}{2} \right\rceil, 2p - h \right].$$

For this L , choose the $L-p$ free bits $b_1, \dots, b_{L-p}$ such that their sum is exactly $S = 2p-L-h$. This is possible because S is an integer between 0 and $L-p$.

By theorem 5.2, this choice of $L-p$ free bits, together with the p bits fixed by the target v , uniquely determines a valid walk of length L in $C(p) \cong B(2, p)$ starting from u . Applying the Structural Path Correspondence (theorem 2.1) yields a walk of length exactly $2p$ in G_p .

Hence, for every $u, v \in V_p$, there exists a walk of length exactly $2p$ in G_p . Thus,

$$\exp(G_p) \leq 2p.$$

3. Lower Bound ($\exp(G_p) > 2p-1$). We exhibit a pair of vertices (u, v) for which no walk of length exactly $2p-1$ exists in G_p .

Let

$$u = 2^p - 1, \quad \text{so that } \Phi_p(u) = (1, 1, \dots, 1)_2,$$

and let v be any odd vertex. We will show that there exists no walk of length exactly $2p-1$ from u to v in G_p .

Suppose, for contradiction, that there exists a walk in G_p from u to v of length exactly $2p-1$. Let L be the length of the corresponding underlying walk in $C(p) \cong B(2, p)$.

Case 1: $L < p$. Since the initial word is $(1, 1, \dots, 1)$, after $L < p$ shifts, the register still contains $p-L$ of the original 1s. As these bits are shifted rightward, every shifted-out bit during these L transitions is 1. Hence, all L source vertices $w_0, \dots, w_{L-1}$ of these transitions are odd. By the Structural Path Correspondence (theorem 2.1), each of the L transitions contributes exactly 2 steps in G_p . Thus,

$$\ell_{G_p}(u, v) = 2L,$$

which is even. Since $2p-1$ is odd, this case is impossible.

Case 2: $L \geq p$. Using the length formula established in Part 1, since $H(\Phi_p(u)) = p$, we have

$$\ell_{G_p}(u, v) = L + p + S,$$

where $S \geq 0$. For this length to equal $2p-1$, we would need

$$L + p + S = 2p - 1 \iff L + S = p - 1.$$

But $L \geq p$ and $S \geq 0$ imply $L + S \geq p$, a contradiction.

Both cases are impossible. Therefore, no walk of length $2p-1$ exists from $u = 2^p - 1$ to any odd vertex v . In particular, the entry $(A^{2p-1})_{u,v}$ is zero for all odd v .

Hence,

$$\exp(G_p) > 2p - 1.$$

Combining the upper bound $\exp(G_p) \leq 2p$ and the lower bound $\exp(G_p) > 2p-1$, we conclude that

$$\exp(G_p) = 2p.$$

□

6 Limits of Modular Dynamics and Transition to Infinity

6.1 Loss of Binary Branching in the Infinite Limit

The inverse limit of the finite modular systems

$$\mathbb{Z}/2^p\mathbb{Z}$$

is not the set of natural numbers but the ring of 2-adic integers

$$\mathbb{Z}_2 = \varprojlim_{p \rightarrow \infty} \mathbb{Z}/2^p\mathbb{Z}.$$

On $\mathbb{Z}_2$, the Bernstein–Lagarias conjugacy remains valid and the modular Syracuse map is still conjugate to the one-sided binary shift. Consequently, the De Bruijn structure persists completely in the 2-adic setting: at each step, one may still interpret the dynamics as the injection of a binary digit into an infinite register.

However, the classical Collatz dynamics are defined on the embedded subset

$$\mathbb{N} \subset \mathbb{Z}_2.$$

When the dynamics are restricted to $\mathbb{N}$, one of the two modular branches generated by division by 2 disappears. Indeed, for an even vertex x , the modular successors are

$$\frac{x}{2} \quad \text{and} \quad \frac{x}{2} + 2^{p-1}.$$

As $p \rightarrow \infty$, the second branch satisfies

$$\frac{x}{2} + 2^{p-1} \longrightarrow +\infty$$

in the usual Archimedean topology and therefore has no finite representative in $\mathbb{N}$.

As a consequence, the De Bruijn branching structure present in the modular graphs G_p is lost in the classical Collatz system. More precisely, the following properties disappear when passing from the finite modular models to the dynamics on $\mathbb{N}$:

- the ability to inject arbitrary bits at each step,
- the strong connectivity of the state space,
- the uniform primitivity property

$$\exp(G_p) = 2p,$$

- the finite-state binary automaton structure.

The resulting dynamics become a deterministic functional system:

$$T(x) = \begin{cases} x/2, & x \text{ even,} \\ 3x + 1, & x \text{ odd.} \end{cases}$$

Therefore, although every finite modular graph G_p is strongly connected, primitive, and admits complete binary controllability through its De Bruijn structure, these properties do not survive the restriction from the 2-adic system $\mathbb{Z}_2$ to the classical dynamical system on $\mathbb{N}$.

6.2 2-Adic Perspective and Relation with Tao's Results

The natural setting for extending the Collatz dynamics beyond the finite modular systems is the ring of 2-adic integers $\mathbb{Z}_2$, which arises as the inverse limit

$$\mathbb{Z}_2 = \varprojlim_{p \rightarrow \infty} \mathbb{Z}/2^p\mathbb{Z}.$$

Within this framework, the Collatz map admits a natural 2-adic extension and remains conjugate to the one-sided binary shift through the Bernstein–Lagarias correspondence. This provides a setting in which probabilistic, ergodic, and harmonic-analytic techniques become available.

It is precisely in this broader 2-adic perspective that the work of [3] is situated. Rather than studying the finite modular graphs individually, Tao analyzes statistical properties of Collatz orbits using methods from analytic number theory and dynamical systems on large scales.

It is important to note that Tao's result is probabilistic and measure-theoretic: it establishes that the set of integers whose Collatz orbit does not eventually fall below a certain bound has density zero. By contrast, the present work establishes exact graph-theoretic connectivity properties for finite moduli. These two frameworks are complementary: the finite controllability proven here is a local, algebraic property, whereas Tao's analysis captures the global statistical behavior. The obstruction to proving the full conjecture lies precisely in the gap between these two regimes: the loss of branching on $\mathbb{N}$ prevents the finite-state controllability from being propagated to the infinite setting.

A fundamental obstacle to proving the classical Collatz conjecture is that the set of natural numbers

$$\mathbb{N} \subset \mathbb{Z}_2$$

forms a highly non-generic and rigid subset of the 2-adic space. Although the finite modular graphs G_p are strongly connected and possess the uniform primitivity property

$$\exp(G_p) = 2p,$$

these finite-state connectivity properties do not imply global convergence of all trajectories in $\mathbb{N}$.

Consequently, the universal convergence of every positive integer to the cycle

$$1 \rightarrow 4 \rightarrow 2 \rightarrow 1$$

cannot be deduced merely by passing to the limit $p \rightarrow \infty$ in the modular models. The transition from the controllable finite De Bruijn structures to the deterministic dynamics on $\mathbb{N}$ introduces arithmetic constraints that are not captured by finite-state connectivity alone.

7 Conclusion

We have shown that the modular Syracuse graph G_p is a strongly connected primitive digraph whose structure is obtained from the binary De Bruijn graph through a local path rerouting process.

This correspondence yields exact formulas

$$\text{diam}(G_p) = 2p - 1, \quad \exp(G_p) = 2p,$$

for every $p \geq 2$.

The results reveal a complete finite-state dynamical structure behind the modular Collatz system. However, the transition from the inverse limit $\mathbb{Z}_2$ to the classical dynamics on $\mathbb{N}$ destroys the branching mechanism responsible for these connectivity properties, leaving the global Collatz conjecture open.

A Exhaustive Numerical Verification for $p = 3$ ($M = 8$)

We consider the modular Syracuse graph

$$G_3 = (V_3, E_3), \quad V_3 = \{0, 1, 2, 3, 4, 5, 6, 7\}.$$

This example illustrates the structural correspondence with the binary De Bruijn graph $B(2, 3)$, as well as the Diameter and Primitivity Exponent Theorems.

A.1 Calculated Transitions of G_3

The odd vertices follow the Syracuse rule

$$x \mapsto (3x + 1) \bmod 8,$$

while the even vertices have the two successors prescribed by modular division by 2.

- **Odd vertices**

$$1 \rightarrow 4, \quad 3 \rightarrow 2, \quad 5 \rightarrow 0, \quad 7 \rightarrow 6.$$

- **Even vertices**

$$0 \rightarrow \{0, 4\}, \quad 2 \rightarrow \{1, 5\}, \quad 4 \rightarrow \{2, 6\}, \quad 6 \rightarrow \{3, 7\}.$$

The complete adjacency table is therefore

x	$\Gamma^+(x)$
0	$\{0, 4\}$
1	$\{4\}$
2	$\{1, 5\}$
3	$\{2\}$
4	$\{2, 6\}$
5	$\{0\}$
6	$\{3, 7\}$
7	$\{6\}$

A.2 The Modular Syracuse Graph G_3

Below is the visual structure of G_3 . Even nodes have an out-degree of 2 (represented by black edges), and odd nodes have an out-degree of 1 (represented by red edges).

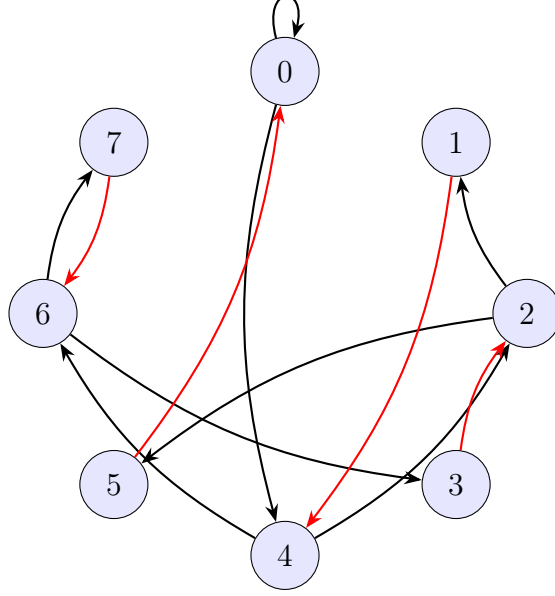


Figure 1: The modular Syracuse graph G_3 on $\mathbb{Z}/8\mathbb{Z}$. Even vertices have out-degree 2 (black edges), while odd vertices have out-degree 1 (red edges).

A.3 Adjacency Matrix A of G_3

$$A = \begin{pmatrix} 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \end{pmatrix}.$$

A.4 Diameter Verification (theorem 4.1)

For $p = 3$, theorem 4.1 predicts

$$\text{diam}(G_3) = 2p - 1 = 5.$$

Consider the vertices

$$u = 7, \quad v = 0.$$

A shortest path is

$$7 \rightarrow 6 \rightarrow 3 \rightarrow 2 \rightarrow 5 \rightarrow 0,$$

which has length 5.

Under the Bernstein–Lagarias conjugacy [1], this path corresponds to the De Bruijn sequence

$$(1, 1, 1)_2 \rightarrow (0, 1, 1)_2 \rightarrow (0, 0, 1)_2 \rightarrow (0, 0, 0)_2.$$

More precisely,

-

$$\Phi_3^{-1}((1, 1, 1)_2) = 7,$$

producing the subdivided transition

$$7 \rightarrow 6 \rightarrow 3$$

of length 2;

-

$$\Phi_3^{-1}((0, 1, 1)_2) = 3,$$

producing the subdivided transition

$$3 \rightarrow 2 \rightarrow 5$$

of length 2;

-

$$\Phi_3^{-1}((0, 0, 1)_2) = 5,$$

with

$$3 \cdot 5 + 1 \equiv 0 \pmod{8},$$

yielding the direct transition

$$5 \rightarrow 0$$

of length 1.

The total length is therefore

$$2 + 2 + 1 = 5,$$

which agrees with

$$\text{diam}(G_3) = 5.$$

A.5 Primitivity Exponent Verification (theorem 5.1)

For $p = 3$, theorem 5.1 predicts

$$\exp(G_3) = 2p = 6.$$

This prediction is confirmed by the powers of the adjacency matrix A .

- At step 5, the matrix A^5 still contains zero entries. In particular, there is no walk of length 5 from vertex 7 to any odd vertex

$$\{1, 3, 5, 7\}.$$

Hence

$$\exp(G_3) > 5.$$

- At step 6, every entry of A^6 is strictly positive:

$$A^6 > 0.$$

Therefore every ordered pair of vertices (u, v) is connected by a walk of length exactly 6.

Consequently,

$$\exp(G_3) = 6,$$

in perfect agreement with theorem 5.1.

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